6.6 First decimals: activities

← Unit overview

1. Coin decimals

You need: €1 coins, 10 cent coins, 1 cent coins (or 1 and 2 cent coins, or paper slips).

1. "€1 is 1 whole. How many 10 cent coins make €1?" (10.) "So one 10 cent coin is one tenth of a euro: €0,10."
2. "How many 1 cent coins make €1?" (100.) "One cent is one hundredth: €0,01."
3. Build amounts: "Show me €2,45." (2 euros, 4 ten-cent coins, 5 one-cent coins.) Write it in a table:

```
wholes (euros) | , | tenths (10 cent) | hundredths (1 cent)
      2        | , |        4         |         5          →  €2,45
```

4. Swap: he builds an amount, you write it.

Talk about it: "Why is €2,05 different from €2,50?"

2. Metre stick

You need: a tape measure or metre stick, sticky notes.

1. One metre = one whole. Ten centimetres = one tenth of a metre (0,1 m). One centimetre = one hundredth (0,01 m).
2. Measure his height: 1 m and 42 cm → 1,42 m. Measure family members, a table, a door.
3. Write each one both ways: "1 m 42 cm = 1,42 m".
4. Try the tricky one: something about 2 m 8 cm long. How do you write it? (2,08 m.)

Talk about it: "Who is taller: someone who is 1,5 m or someone who is 1,45 m?"

3. Zoom-in number line

You need: a long strip of paper, a pen.

1. Draw a line from 0 to 1. Mark the tenths: 0,1, 0,2 … 0,9.
2. "Where is 0,35?" It's between 0,3 and 0,4. Draw a magnifying-glass circle around that bit and a second, bigger line underneath from 0,3 to 0,4, with ten small steps: 0,31, 0,32 … 0,39.
3. Place 0,35 exactly.

Talk about it: "Are there numbers between 0,35 and 0,36?" (Yes! 0,355, for example. That's a good thing to wonder about.)

4. Sports results

You need: a tape measure, a stopwatch (only to *measure*, never to rush him), an outdoor space.

Hold a family long-jump or standing-jump contest. Measure each jump in metres and centimetres, write it as a decimal (1,63 m), and make a results table. Who won? By how many centimetres?

You can also look at real results online: athletics, swimming or speed skating (*schaatsen*) times often use hundredths of a second.

Talk about it: "Why do sports results use hundredths?"

5. Price tags

You need: sticky notes, a few household items, a supermarket folder.

Price items around the house (€0,85, €1,20, €1,05, €0,09). He puts them in order from cheapest to most expensive. Then use the folder: find the cheapest and most expensive item on one page.

Talk about it: “Which was the hardest pair to compare? Why?”

Conversation prompts (any time)

- “Where do you see a comma in a number in real life?” (Prices, petrol, temperatures, sports, body temperature 37,5 °C.)
- “Which is closer to 1: 0,9 or 0,99?”
- “Is 0,5 the same as 1/2? How do you know?”
- “Can you think of a number between 1,1 and 1,2?”